

SU_HACKATHON 2026

Engineering & Manufacturing

TexNova

Problem Statement

AI and Automation for Manufacturing SMEs

TEAM

TexNova

TEAM ID

TEM105

MEMBERS

5 Members

Industry 4.0 • Affordable • Built for Indian MSMEs

AI and Automation for Manufacturing SMEs

THE GROUND REALITY

ROOT CAUSES

850+

manufacturing units in Bhilwara (textile, yarn, fabrics)

500+

Serviceable Units (>10 machines)

₹20K–60K

cost per hour of downtime for medium units

15-20 hrs

average unplanned downtime per month per unit

₹240 Cr+

Annual Value Creation Opportunity
 $500 \times 10 \times 40,000 \times 12 = ₹240 \text{ Crore annual loss}$



No real-time production visibility



Manual monitoring leads to human error



Breakdowns happen without warning



Quality issues detected too late



Scheduling based on guesswork

SmartSME: Smart Monitoring Solution for Textile Machines

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OUR SOLUTION

Key Features

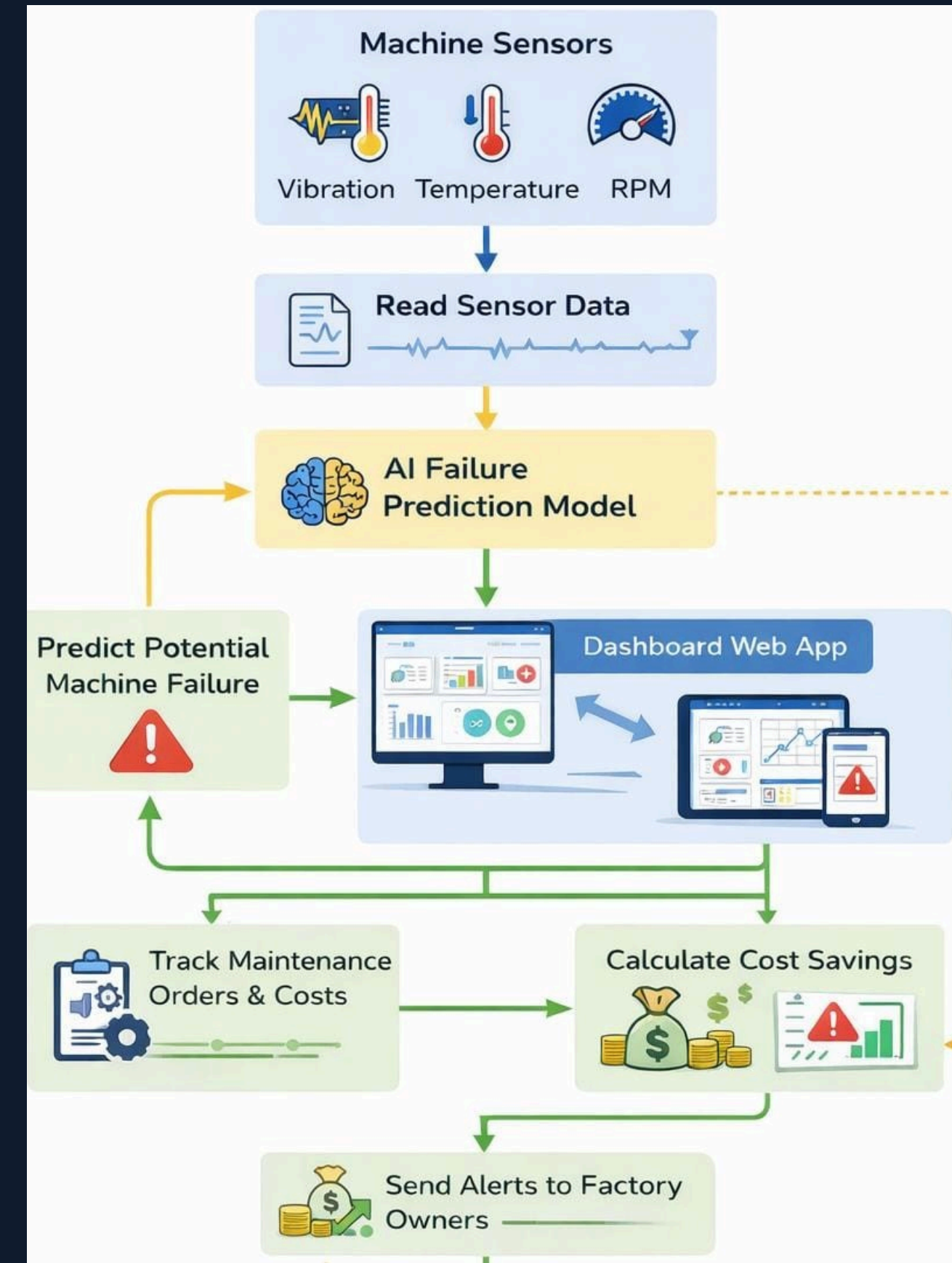
- Real-time machine monitoring
- Early alert before machine failure
- Schedule dataf all the Machine Systematically
- Mobile dashboard for factory owners

Workflow Optimization using SmartSME

- Sensors collect real-time machine data
- System detects abnormal machine behavior
- Alert is sent to operator via mobile app
- Maintenance is done before failure occurs

Tech Stack

- Python, HTML, CSS, JS
- Fast API
- AI Models:SVM,Decision Tree, Random Forest Logistic Regression
- Numpy,Pandas,Tensorflow



Dashboard

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OUR SOLUTION

It will Show :

- How many machines are there?
- How many machines are shown healthy, warning, critical condition
- Hourly risk

For Example:

A Healthy Power Loom shows:

- Temperature: 47°C
- RPM: 420
- Vibration :1.4 mm/sec

Real Time AI Monitoring

It will Show :

- Live sensed data of any Machine
- 30- Day Trend Chart



AI Prediction

Machine failure Probability

Show failure probability

- Healthy
 - Low Risk
 - Moderate Risk
 - Crictical Risk
- According to:
- Range shows
- Vibration
 - Temperature
 - RPM

It will show Real- time Model accuracy



Machine Life- Cycle And Maintenance

Description of Machine Issues

1

New
≤ 50 min wear
Brand new.
Minimal wear.
All sensors in optimal range.
0% failure risk

2

Early Use
≤ 100 min wear
Breaking in.
Minor wear patterns forming.
Low risk.
2–5% failure risk

3

Mid-life
≤ 150 min wear
Normal wear.
Monitor vibration.
Schedule service soon.
8–15% failure risk

4

Mature
≤ 200 min wear
High wear.
Vibration rising.
Plan replacement parts.
20–40% failure risk

5

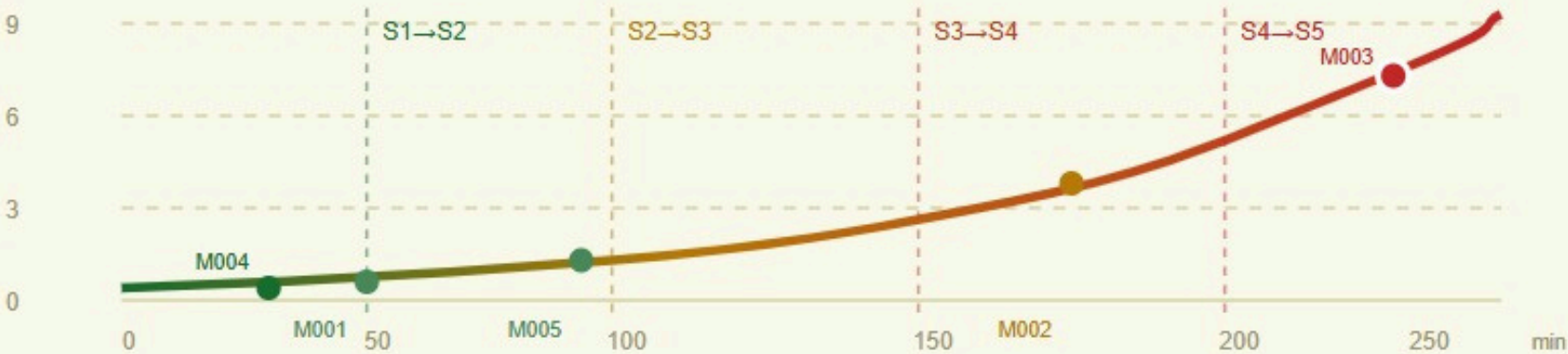
End-of-Life
> 200 min wear
Critical wear.
Replace tool immediately.
High failure risk.
>60% failure risk

For Example:

Maintenance

- It can be used by technicians and owner
- will show, open and recent jobs and satus of severity
- how much will it cost to fix the fault

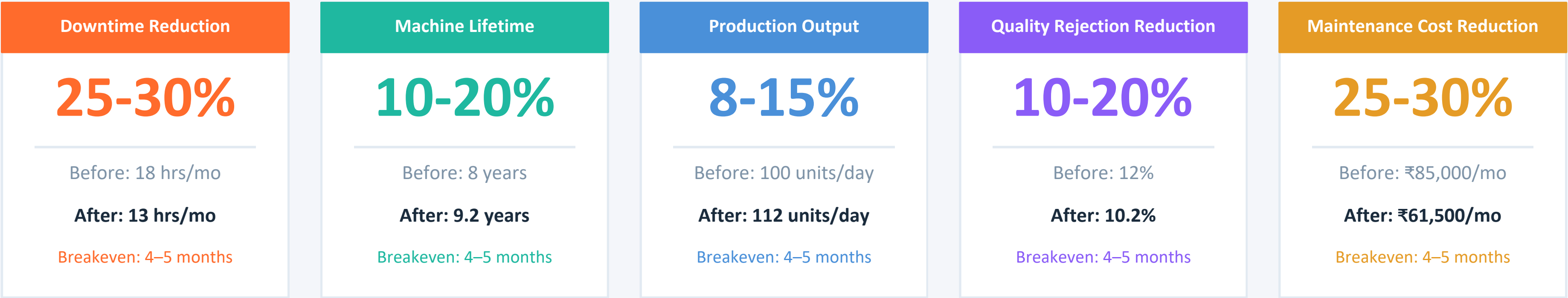
Vibration Trend by Lifecycle Stage



From Bhilwara to Bharat: Transforming MSME Manufacturing

Measurable ROI • Proven Business Model • Pan-India Vision

IMPACT, VIABILITY & SCALE



Monetising

- Rs. 199/month - 3 machines, failure prediction, basic alerts.
- Rs. 1999/month - upto 10 machines, life cycle predction, all alerts, reports.
- Rs. 4999/month - unlimited machines, advance analytics, future features.

Advertising

- Industrial exhibition.
- publication by Industrial Marketing companies like: India Mart, Trade India, Moglix, Shakedeal, BulkMRO.
- With the help of installation of hardware.

"We're building the Operating System for Indian Manufacturing — starting Bhilwara, scaling to Bharat."